VIMAL JYOTHI ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to APJ Abdul Kalam Technological University)

B.TECH DEGREE S2 (R/S) EXAMINATION APRIL 2026 (2024 SCHEME)

GXEST203 FOUNDATIONS OF COMPUTING: FROM HARDWARE ESSENTIALS TO WEB DESIGN

(CC, CS, CU, CN, AD, EC)

Max. Marks : 60 **Duration :2 Hours 30 Minutes**

SCHEME OF VALUATION / ANSWER KEY

PART A			
S. No.	**Answer all questions. Each question carries 3 marks**	**CO**	**Marks**
1	MAC Address – Purpose (3 Marks) ● Definition of MAC→ 1 mark ● Purpose of a MAC address in a NIC → 1 mark ● Role in data link layer / frame delivery → 1 mark	CO1	(3)
2	Firmware (3 Marks) ● Definition → 1 mark ● Function→ 1 mark ● Storage: ROM / EEPROM / Flash → 1 mark	CO1	(3)
3	LMC Data Movement Instructions (3 Marks) ● Correct identification of instructions → 2 marks (LDA, STA, INP, OUT – any relevant) ● Explanation (moves data between memory/accumulator/I-O) → 1 mark	CO2	(3)
4	Number System Bases (3 Marks) ● Binary = 2 → 0.5 mark ● Decimal = 10 → 0.5 mark ● Octal = 8 → 1 mark ● Hexadecimal = 16 → 1 mark	CO2	(3)
5	ls Command (3 Marks) ● Purpose: List directory contents → 1 mark ● Any two options with meaning → 2 marks ● (e.g., -l, -a, -h, -R)	CO3	(3)
6	DHCP (3 Marks) ● Definition → 1 mark ● Importance (any two points) → 2 marks	CO3	(3)
7	Web Content Creation (3 Marks) ● Any three valid elements → 3 × 1 mark (clarity, structure,readability, SEO, usability)	CO4	(3)

8	CSS Application Methods (3 Marks) ● Inline CSS → 1 mark ● Internal CSS → 1 mark ● External CSS → 1 mark	CO4	(3)

PART B **(Answer any one full question from each module, each question carries 9 marks)**			
Module 1			
9	(a) Device Management Functions (5 Marks) ● Definition of device management → 1 mark ● Device allocation & deallocation → 1 mark ● I/O scheduling → 1 mark ● Buffering, caching, spooling → 1 mark ● Device drivers & interrupt handling → 1 mark (b) I/O Controllers (4 Marks) ● Definition of I/O controller → 1 mark ● Role (interface between CPU & devices) → 1 mark ● Working (data transfer, control signals) → 1 mark ● Example (keyboard controller, disk controller, USB controller) → 1 mark	CO1	(9)
OR			
10	(a) HDD Structure & Working (5 Marks) ● Components: platters, spindle, read/write head → 2 marks ● Tracks, sectors explanation → 1 mark ● Working principle (magnetic storage, rotation, head movement) → 2 marks (b) Optical Drives vs HDD & SSD (4 Marks) ● Optical drive working (laser-based) → 1 mark ● HDD vs Optical (speed, durability) → 1 mark ● SSD vs Optical (no moving parts, faster) → 1 mark ● Any one comparison point (cost/capacity/use) → 1 mark	CO1	(9)
Module 2			
11	(a) Number Conversion & Ordering (5 Marks) ● Correct conversion of each number → 3 marks ● Correct decimal values → 1 mark ● Correct ascending order → 1 mark (b) Base Conversion Error (4 Marks) ● Mistake identified (remainders not reversed) → 2 marks ● Correct procedure explanation → 2 marks	CO2	(9)
OR			

12	Fetch-Execute Cycle (9 Marks) ● Definition of instruction cycle → 1 mark ● Fetch step (PC → MAR → MDR → IR) → 2 marks ● Decode step → 1 mark ● Execute step (LOAD operation) → 2 marks ● Register involvement (PC, MAR, MDR, IR, ACC) → 2 marks ● Diagram / sequence clarity → 1 mark	CO2	(9)
	Module 3		
13	(a) DHCP DORA Process (5 Marks) ● Discovery → 1 mark ● Offer → 1 mark ● Request → 1 mark ● Acknowledgement → 1 mark ● Example explanation → 1 mark (b) DHCP Issues (4 Marks) ● Any three issues → 3 marks (IP conflict, server down, pool exhaustion, misconfiguration) ● Solutions → 1 mark	CO3	(9)
	OR		
14	Network Topologies (9 Marks) ● Explanation of types (bus, star, ring, mesh) → 4 marks ● Advantages → 2 marks ● Disadvantages → 2 marks ● Real-world applications → 1 mark	CO3	(9)
	Module 4		
15	(a) CSS Styling (5 Marks) ● Background styling → 1 mark ● Heading style → 1 mark ● Font styling → 1 mark ● Correct syntax → 1 mark ● Overall structure → 1 mark (b) External CSS Advantages (4 Marks) ● Reusability → 1 mark ● Maintainability → 1 mark	CO4	(9)

	● Performance (caching) → 1 mark ● Separation of content & design → 1 mark		
	OR		
16	(a) FTP Role (5 Marks) ● Definition of FTP → 1 mark ● File transfer function → 1 mark ● Upload/download management → 1 mark ● Directory management → 1 mark ● Efficiency explanation → 1 mark (b) FTP Upload Steps (4 Marks) ● Connect to server → 1 mark ● Authentication → 1 mark ● File transfer (upload) → 1 mark ● Verification → 1 mark	CO4	(9)